

Lecture 15:-

Joint and Marginal Probabilities

Continue ~~with~~ with the last lecture

How about 2 random variables?

"Bag i has i blue balls and 2 green balls."

let $i=6$

Random variables: B = bag picked
 C = color of ball picked '1' = blue, '2' = green "number assigns to colours".

B \ C	$P(B=b, C=c)$		$P(C)$
	$C=1$	$C=2$	
$B=1$	$\frac{1}{6} \cdot \frac{1}{3} = \frac{1}{18}$	$\frac{1}{6} \cdot \frac{2}{3} = \frac{2}{18}$	$\frac{1}{6}$
$B=2$	$\frac{1}{6} \cdot \frac{2}{4} = \frac{2}{24}$	$\frac{1}{6} \cdot \frac{2}{4} = \frac{2}{24}$	$\frac{1}{6}$
$B=3$	$\frac{1}{6} \cdot \frac{3}{5} = \frac{3}{30}$	$\frac{1}{6} \cdot \frac{2}{5} = \frac{2}{30}$	$\frac{1}{6}$
$B=4$	$\frac{1}{6} \cdot \frac{4}{6} = \frac{4}{36}$	$\frac{1}{6} \cdot \frac{2}{6} = \frac{2}{36}$	$\frac{1}{6}$
$B=5$	$\frac{1}{6} \cdot \frac{5}{7} = \frac{5}{42}$	$\frac{1}{6} \cdot \frac{2}{7} = \frac{2}{42}$	$\frac{1}{6}$
$B=6$	$\frac{1}{6} \cdot \frac{6}{8} = \frac{6}{48}$	$\frac{1}{6} \cdot \frac{2}{8} = \frac{2}{48}$	$\frac{1}{6}$
$P(C=C)$	0.594	0.405	

As you can see blue ball ($C=1$) is changing with b whereas green ball ($C=2$) always remain constant or 2

Another Way of representation

A \ B		
	1	2
1		
2		
3		

Marginal Probability of B $\leftarrow P(B=1)$ $P(B=2)$ $\leftarrow$ Joint probability distributions of A and B

$\leftarrow$ marginal probability of A

Effect is removed

$$P(B=1) = \sum_i P(B=1, C=i) \quad (\text{same rule of summation of mutually exclusive events})$$

"sum over all possible values of C".

But we already know the $\frac{1}{6}$ for $P(B=1)$!!

Why do we need to do this summation?

In real life we not get all just joint probability and you have to use it to derive other

B	C=1	C=2	$P(B=b)$
B=1	$\frac{1}{8}$	$\frac{2}{18}$	
B=2	$\frac{2}{24}$	$\frac{2}{24}$	
B=3	$\frac{3}{30}$	$\frac{2}{30}$	
B=4	$\frac{4}{36}$	$\frac{2}{36}$	
B=5	$\frac{5}{42}$	$\frac{2}{42}$	
B=6	$\frac{6}{48}$	$\frac{2}{48}$	
$P(C=c)$			

→ Always will be one

When we collect data from real world:

- Sensor (signal + noise)

We have important RVs (signal) and noise mixed together!

We are only able to measure joint probabilities. But we are interested in marginal of one RV.